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School of Electronic Information and Communications
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Profile

Master's student in Electronic Information at Huazhong University of Science and Technology (expected graduation Jun 2027), focusing on **robot motion planning and embodied AI**. My research combines lookahead planning and learned self-collision prediction to support reliable and efficient robotic manipulation. I bring hands-on experience in reinforcement learning, model fine-tuning for spoken-command understanding, simulation-based evaluation, and deploying perception models on robots. I am motivated by the goal of enabling robots to anticipate upcoming actions and adapt their motion as tasks and environments change, supporting **fluent and reliable task execution**.

Education

M.S. in Electronic Information **Sep 2024 – Present**
Huazhong University of Science and Technology, Wuhan, China
Admitted through merit-based recommendation.
GPA: 3.79/4.00; top 5%
Selected coursework: Applied Advanced Engineering Mathematics (97/100); Artificial Intelligence I (91/100).
Expected graduation: Jun 2027

B.S. in Electronics and Information Engineering **Sep 2020 – Jun 2024**
Huazhong University of Science and Technology, Wuhan, China
Qiming Experimental Class (project-based curriculum).
GPA: 3.74/4.00
Selected coursework: Calculus I (99/100); Discrete Mathematics II (93/100); Linear Algebra (89/100).

Research Experience

FWIK: Multi-Waypoint IK Selection via Lookahead Graph Search **Jun 2025 – Sep 2026**
First author and sole student author; led the problem formulation, method design, implementation, experiments; manuscript under review.

- Formulated multi-waypoint IK selection as layered graph search over a finite future window, ranking candidate paths by estimated joint-motion time.
- Used a learned transition model to screen likely self-collisions before MoveIt 2 planning, while retaining geometric trajectory validation in the downstream planner.
- Built a simulation and evaluation pipeline for the Franka Panda and KUKA LBR iiwa 7-DoF arms with four planners. The method achieved 96.8–100% sequence success and 12.8–13.1% lower mean planned trajectory duration than STAMPEDE on completed sequences.

PPO-Based Robot-Arm Motion Planning in Simulation **Sep 2023 – Jun 2024**
Undergraduate Thesis Research

- Built a PyBullet and Stable-Baselines3 pipeline for PPO training on a simulated 6-DoF Z-Arm S622.
- Formulated reaching in chemical laboratory settings as a Markov decision process, with continuous joint-angle actions and rewards for target proximity, orientation, collision avoidance, and motion efficiency.
- Evaluated the best policy for each gripper-to-target distance threshold over 1,000 simulation trials: 92.9% success at 15 mm, 97.7% at 30 mm, and 98.5% at 45 mm.

Internship Experience

Anker Innovations, Embodied-Cognition Agent Department **May 2026 – Aug 2026**
Algorithm Engineering Intern

- Defined structured task categories and key-information extraction rules for spoken robot commands, and built training and evaluation workflows for task-understanding models.
- Independently built training datasets and fine-tuned the model, refining data based on failure-case analysis and evolving product requirements.

Selected Projects

Huawei AP Quality Map Team **Sep 2024 – Apr 2025**

Project Lead, Crowdsourced WiFi AP Quality Assessment

- Led a crowdsourced WiFi access-point quality assessment project; designed evaluation metrics and a data dictionary for dense deployments.
- Built automated terminal-log collection and cleaning workflows and a quality-of-experience (QoE) annotation tool linking subjective user experience with objective measurements for subsequent modeling.

National University of Singapore, School of Computing **Jul 2024 – Aug 2024**

Project Member, Perception-Driven Indoor Cleaning Robot

- Integrated robot hardware, perception models, and motion control in an indoor cleaning prototype that automatically detected crumpled paper and avoided walls and stairs. The project aimed to support indoor mobility for people with visual impairments, including wheelchair users.

Honors and Awards

Second-Class Postgraduate Scholarship, HUST 2025–2026

First-Class Postgraduate Scholarship, HUST 2024–2025

Provincial First Prize, "Datang Cup" 5G+ Innovative Application Simulation Design Competition May 2023

Provincial Second Prize, 13th Chinese Mathematics Competitions (Non-Mathematics) Dec 2021

Scholarship for Self-Improvement, HUST 2021; 2022

Outstanding Individual in Social Practice, HUST 2021; 2022

Leadership and Service

- **English-Language Broadcaster and News Editor** Apr 2021 – Dec 2022
- **Media Team Lead and Outreach Representative, Dian Group** May 2022 – Mar 2024
- **Community Service and Educational Volunteering** (95 hours) Sep 2020 – Jun 2023

Technical Skills

- **Robotics and simulation:** ROS 2, MoveIt 2, motion planning, inverse kinematics, simulation-based evaluation
- **Machine learning and AI:** Reinforcement learning, Transformer models, model fine-tuning and evaluation
- **Programming:** Python, C, JavaScript, HTML, MATLAB
- **Development tools:** Linux, Git, PyTorch, Unity

Additional Information

- **Languages:** English (IELTS 7.5), Mandarin Chinese (Native), French (Basic)
- **Interests:** Choral Singing (Alto) | Graphic Design (Hobbyist) | Robotics | AI for Science
- **Personal Goal:** To design and build my own small robots that address practical problems in daily life.

Referees

References available upon request.